L11H8 is not a previous-token head: it reads nearby countries

GPT-2 small, 200 random rows × 11 countries per filler type. Each row has two versions that differ only at position τ. Δ = attention from the final token T to τ, country − original, per head, at every distance $T - \tau = 1 \ldots 31$.

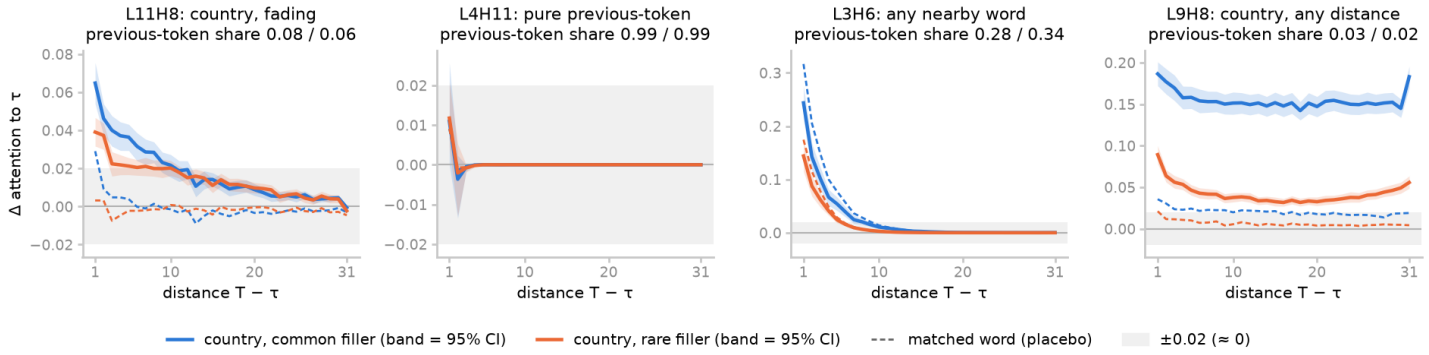
original BOS | she | sc | … 26 random tokens … | age | dis | ious **country** BOS | she | sc | … 26 random tokens … | France | dis | ious (τ = 30, distance 2)

Conclusion (common / rare filler)

- **L11H8 reads countries, and less so the farther away they are.** It gives the previous token only 8% / 6% of its attention on random text. An adjacent country raises its attention to that spot by **+0.065** / **+0.039**, which is 0.036 more than a matched capitalised word. The boost fades to +0.041 / +0.027 at distances 2–4 and about +0.006 at 16–31.
- **Only L4H11 is a clean previous-token head.** It puts at least 98.5% of its attention on the previous token whatever the content, and $\Delta \approx 0$ beyond distance 1. The other Exp 1 "previous-token" heads react to content within 8 tokens: L2H3, L2H9 and L3H6 to any capitalised word-start; L2H5 and L3H8 prefer countries; L1H0 and L2H8 avoid them.
- **Four late heads find the country at any distance**: L9H8, L10H3, L11H7 and L11H2, above control at every distance in both filler types. A 30-row probe suggests L9H8 and L11H2 respond to places, L10H3 to named entities in general, and L11H7 partly to any frequent word.

1 · Four behaviours under one test

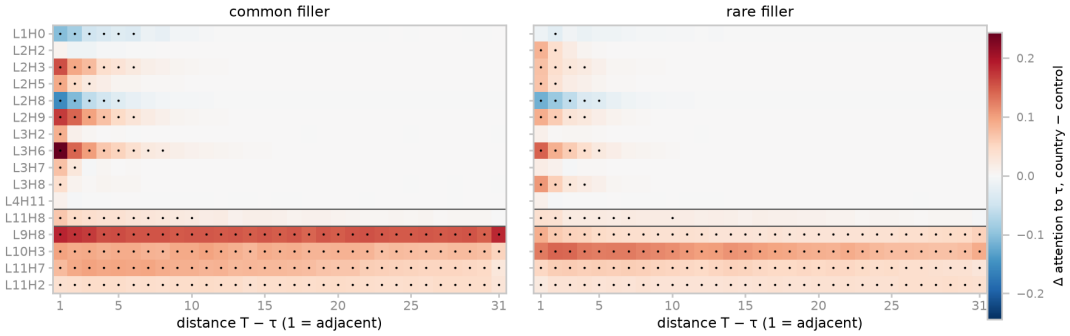
solid = country, dashed = matched capitalised word; grey band = $|\Delta| < 0.02$



L11H8 looked like a previous-token head in Exp 10 only because its attention is otherwise diffuse, so an adjacent country became its top key.

2 · "Previous-token" heads react only nearby

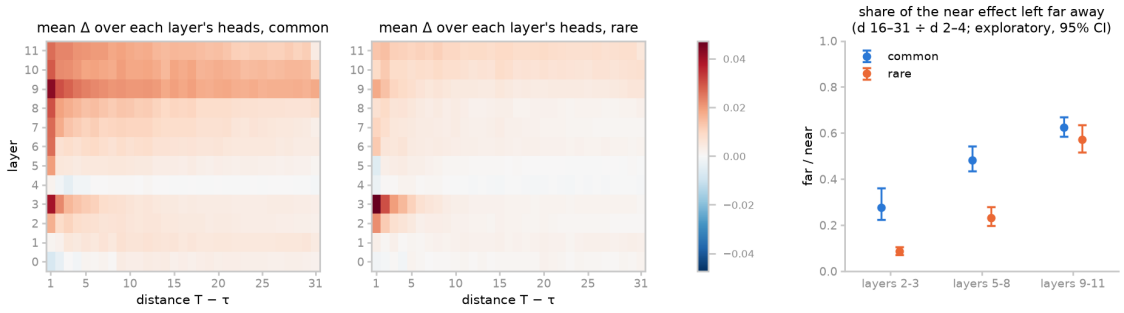
dots = BH $q \leq 0.05$ and $|\Delta| \geq 0.02$



Top: Exp 1's previous-token heads, nothing meaningful past distance 8. Middle: L11H8. Bottom: the four any-distance heads.

3 · Depth and range (stretch)

"layers 9–11 act globally": only through four heads



Layers 2–3 act locally. Layers 9–11 keep about 60% of their near effect past distance 16, but only because of the four heads above. The pre-registered absolute test failed, because 9–11 are larger at every distance. The right panel is exploratory.

Pre-registered (exp4/PREDICTIONS.md): C3–C6 pass. C1, C2 and H-loc fail. All checks pass, and an independent Hugging Face recomputation matched to 6×10^{-6}. The placebo words are not frequency-matched, hence the probe. Interactive: country slider, per-head viewer.